Single-Cell RNA-seq Analysis Report: Global Transcriptomic Profiling and Consensus Cell-Type Annotation

Bioinformatics Analysis Service

2026-08-03

1. Executive Summary

This report describes the global exploration analysis performed on a human intestinal single-cell RNA-seq dataset derived from Crohn's disease patients and healthy controls.

The objective of this analysis was to generate an integrated representation of the cellular landscape, characterize transcriptional heterogeneity across samples, identify major cellular populations, and establish a robust cell-type annotation framework for downstream biological investigations.

The analytical workflow included:

- quality control and preprocessing of single-cell transcriptomic data
- normalization and feature selection

- dimensionality reduction and integrated embedding generation
- correction of sample-associated variation through batch integration
- unsupervised clustering of transcriptional profiles
- consensus cell-type annotation combining reference-based predictions and marker-based biological validation

The analysis identified **14 transcriptionally distinct clusters**, which were further interpreted and consolidated into **6 major biological cell populations** through integrative annotation.

The resulting annotated Seurat object provides a curated representation of the dataset and serves as the reference framework for subsequent compartment-specific analyses.

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2. Experimental Design

The dataset consists of multi-sample human intestinal single-cell RNA-seq profiles generated from patient-derived biopsy samples.

Parameter	Description
Organism	Homo sapiens
Tissue	Intestinal mucosa (lamina propria and intraepithelial compartments)
Disease model	Crohn's disease and healthy controls
Experimental material	Patient-derived intestinal biopsy samples
Sequencing platform	10x Genomics Chromium single-cell RNA-seq
Dataset type	Multi-sample single-cell transcriptomic dataset
Public dataset	GEO GSE157477
Batch identifier	`sample_id`

Cell number summary

Cells were evaluated before and after quality control filtering.

Sample	Condition	Cells before QC	Cells after QC	Retention (%)
GSM4766839	Normal	7498	7457	99.45
GSM4766840	Normal	4110	4025	97.93
GSM4766841	Crohn's disease	5628	5587	99.27
GSM4766842	Crohn's disease	5634	5537	98.28
GSM4766843	Crohn's disease	7638	7630	99.90
GSM4766844	Crohn's disease	7700	7669	99.60

Sample	Condition	Cells before QC	Cells after QC	Retention (%)
GSM4766845	Normal	6827	6674	97.76
GSM4766846	Normal	5370	5293	98.57
GSM4766847	Crohn’s disease	2687	2626	97.73
GSM4766848	Crohn’s disease	4549	4372	96.11
GSM4766849	Crohn’s disease	2434	2403	98.73

Quality control filtering resulted in high cell retention across all samples (96.1%–99.9%), indicating that the applied filtering strategy removed a limited proportion of cells while maintaining the majority of the original dataset.

3. Data Processing Workflow

The single-cell RNA-seq data processing workflow was implemented using a modular pipeline designed to ensure reproducibility, scalability, and compatibility with downstream integrative analyses. The workflow consisted of the following sequential steps:

1- Raw gene-by-cell count matrix import and initial object construction 2- Generation of Seurat object for unified data handling 3- Quality control (QC) filtering of low-quality cells and potential doublets 4- Data normalization and scaling 5- Identification of highly variable genes (HVGs) 6- Principal component analysis (PCA) for dimensionality reduction 7- Batch effect correction and dataset integration using Harmony 8- Non-linear embedding using UMAP for visualization 9- Graph-based clustering of cells in the integrated latent space 10- Reference-based cell type annotation 11- Consensus-based refinement of cell identity assignments across methods

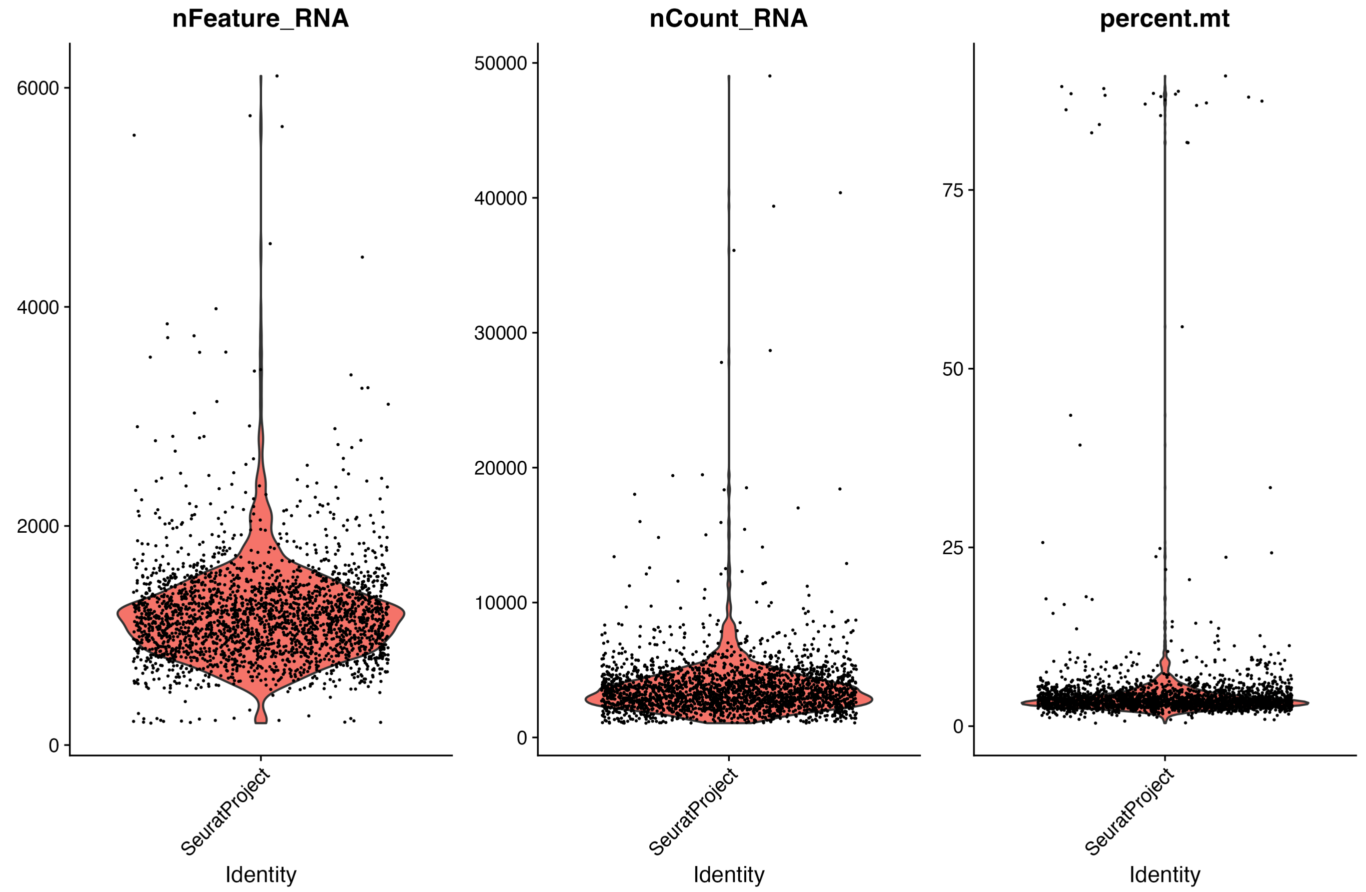
4. Quality Control Assessment

Quality control was performed using:

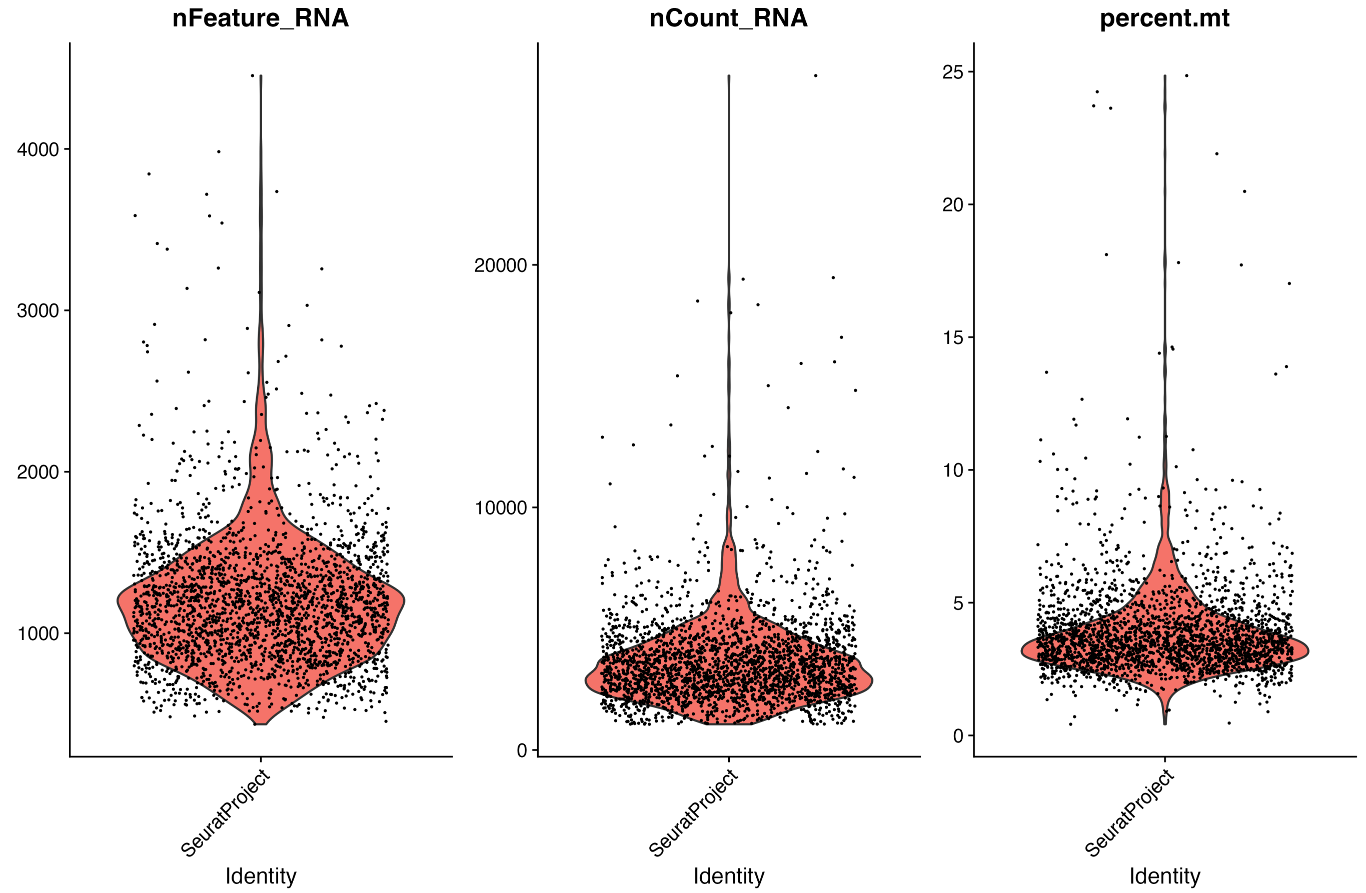
- number of detected genes (nFeature_RNA)
- total transcript counts (nCount_RNA)
- mitochondrial RNA percentage (percent.mt)

4.1 QC metrics distribution

Before Filtering



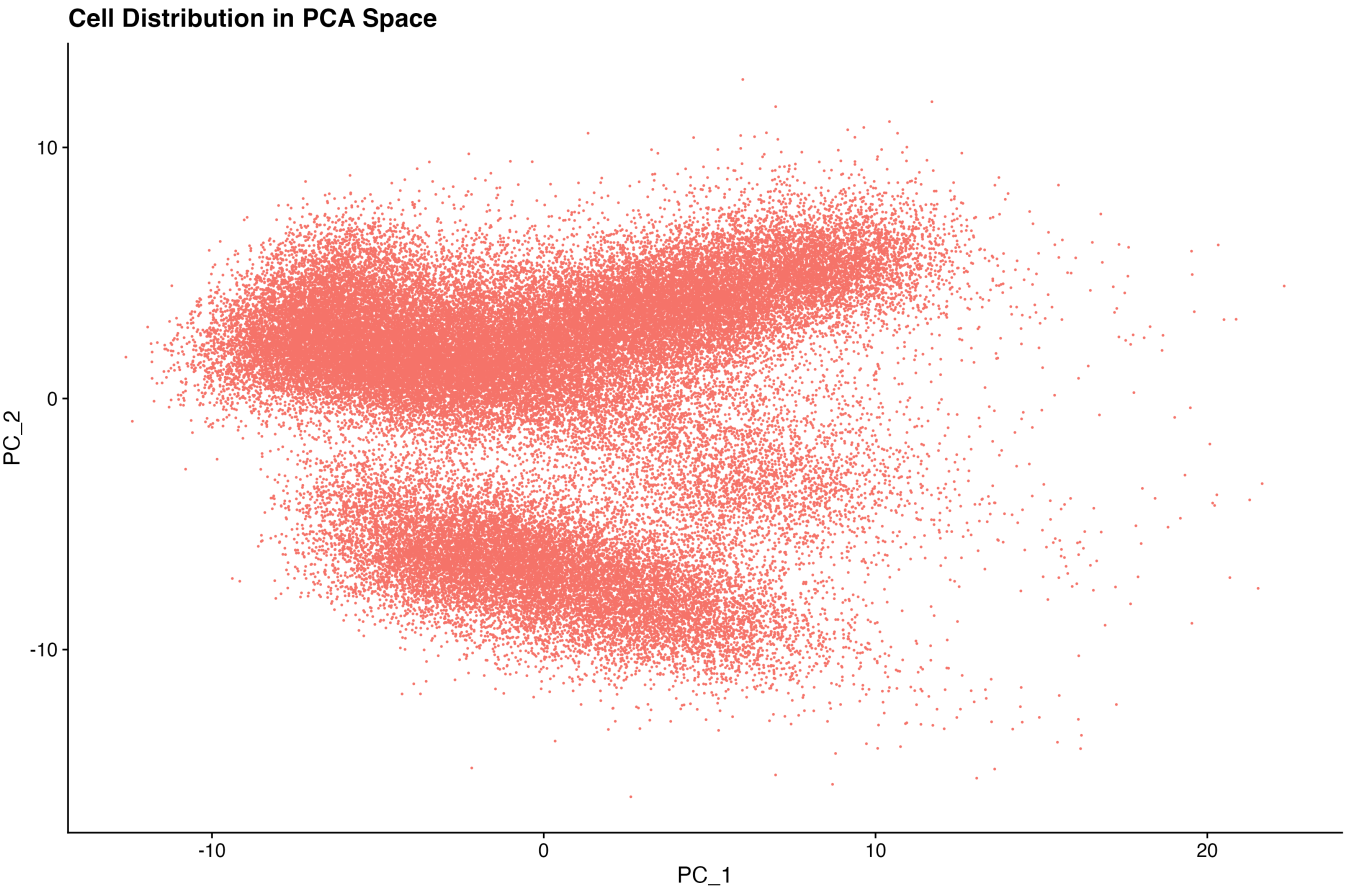
After Filtering



5. Global Transcriptomic Landscape

5.1 Principal Component Analysis

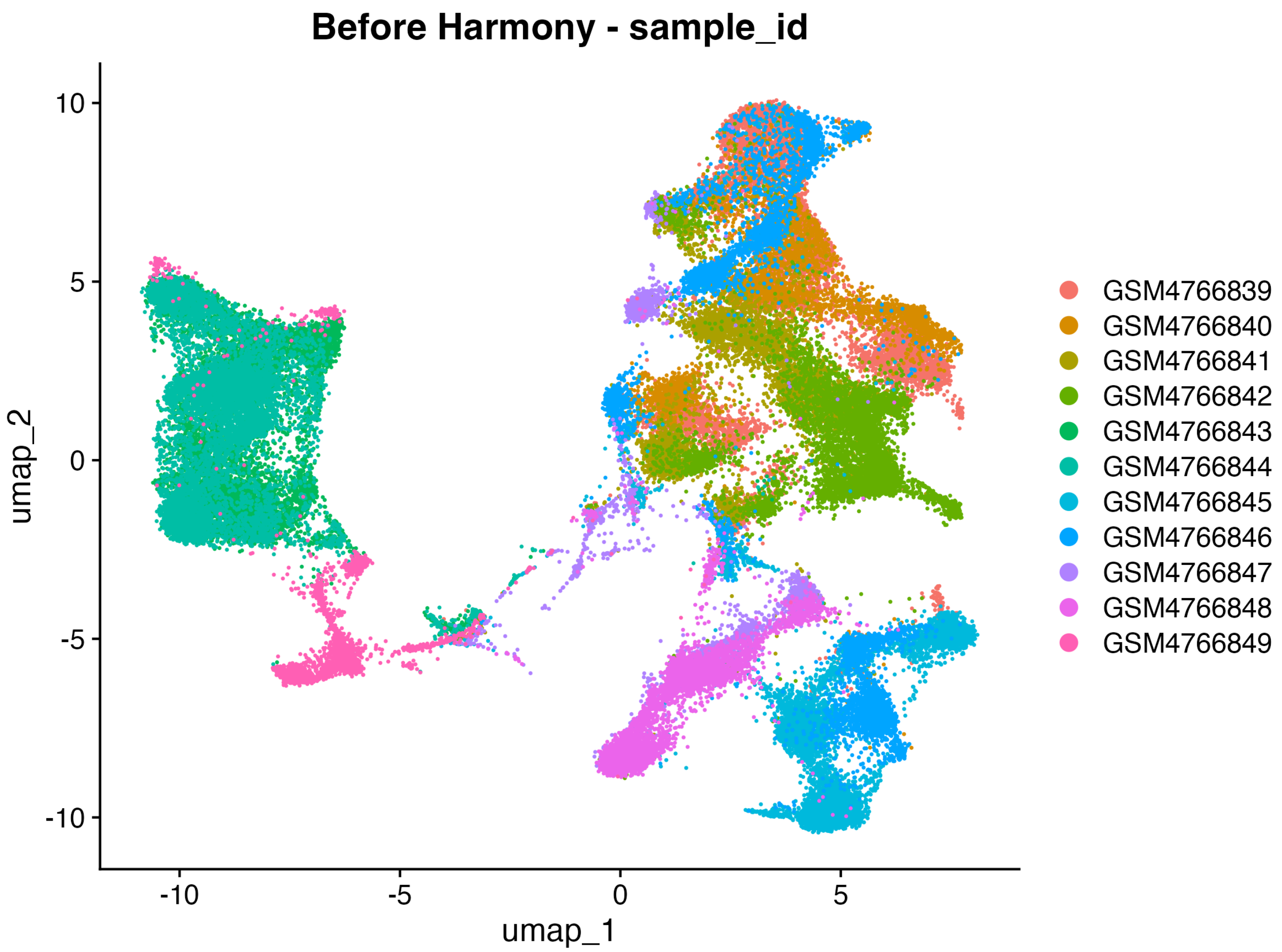
PCA was performed to evaluate major transcriptional sources of variation.



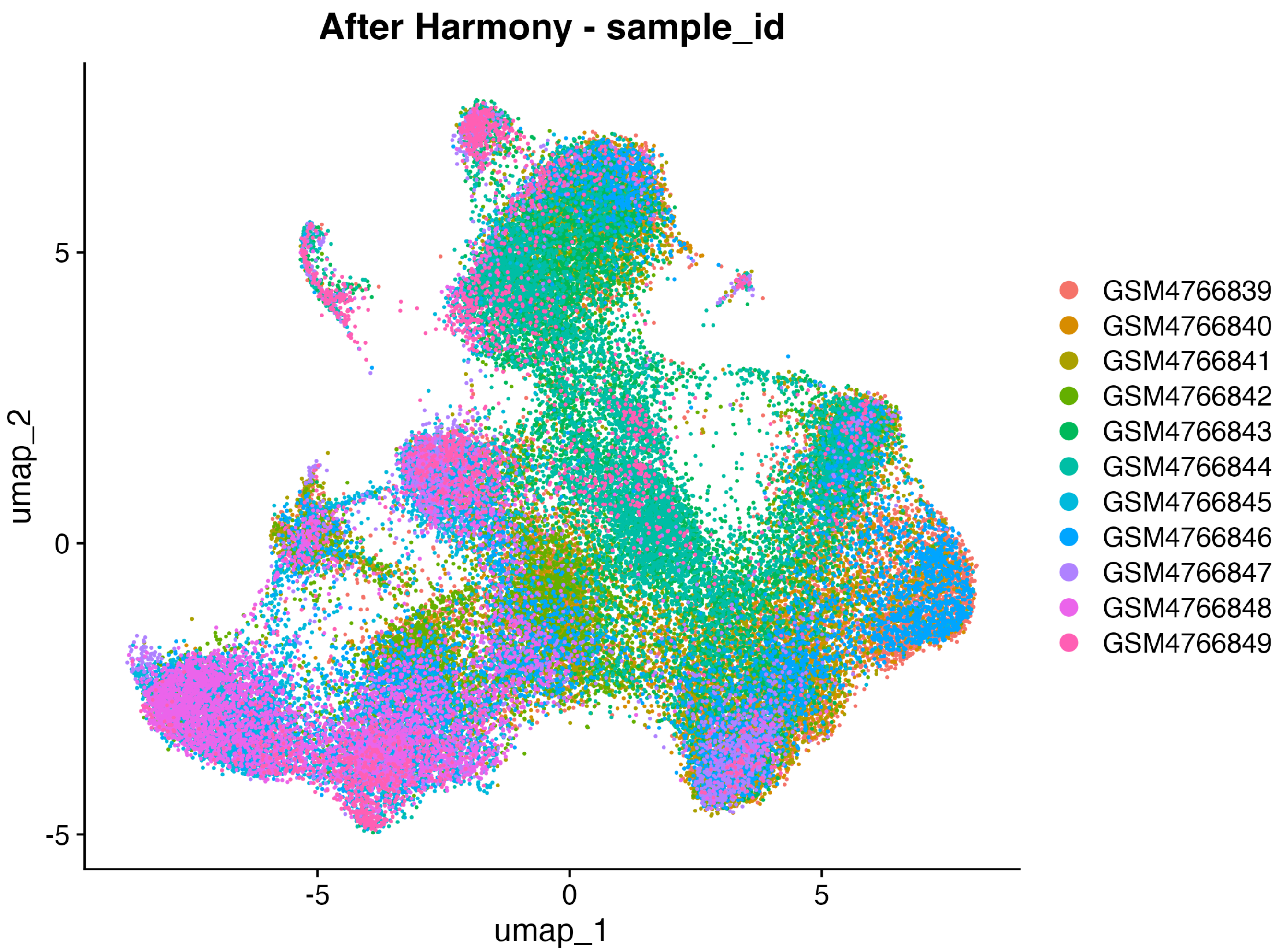
5.2 Harmony Integration

Harmony was applied using sample_id as batch variable to reduce sample-associated variation while preserving biological differences.

Before integration



After integration

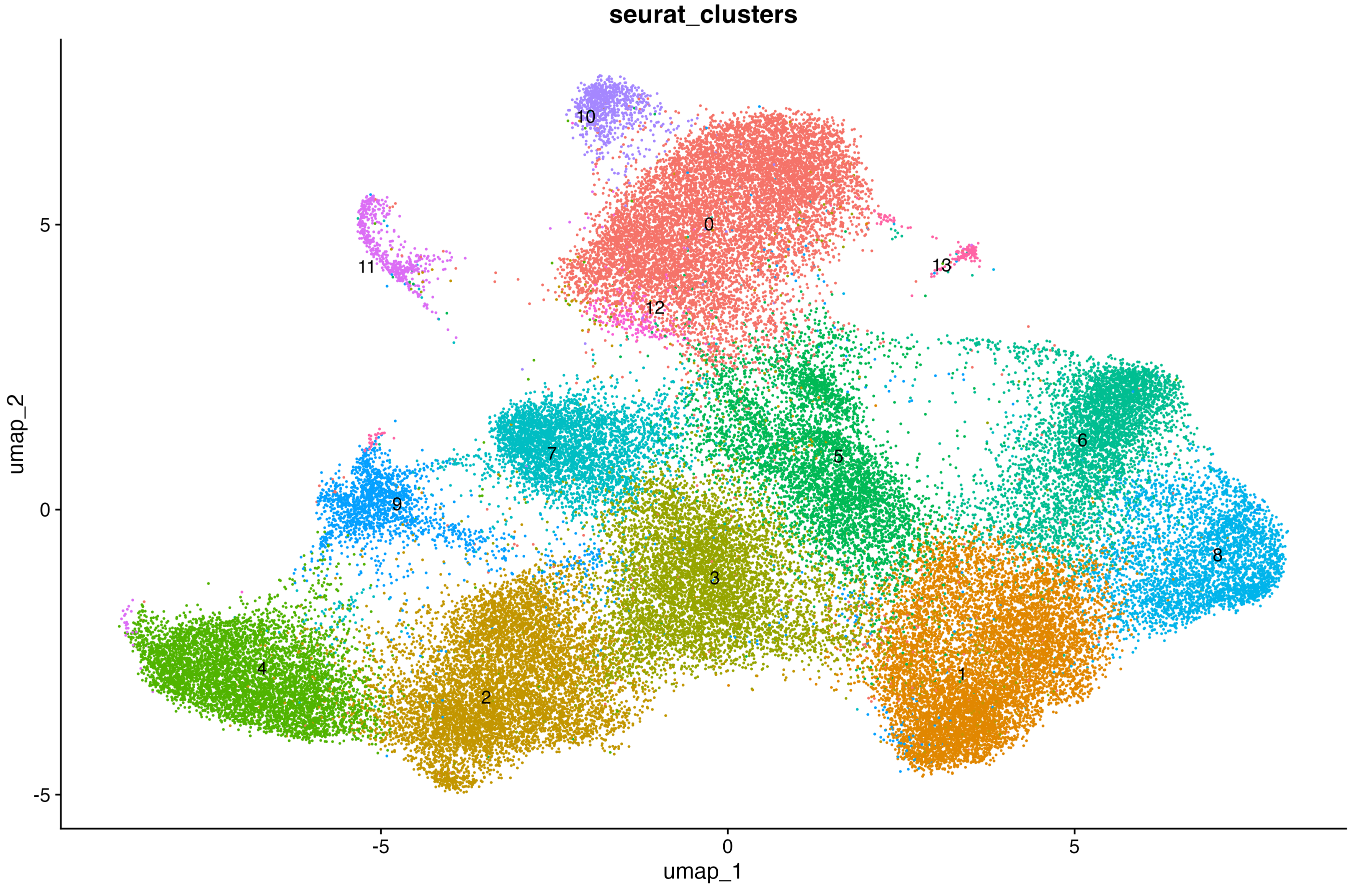


6. Unsupervised Clustering Analysis

Graph-based clustering was performed on the Harmony-integrated latent space using a shared nearest neighbor (SNN) graph.

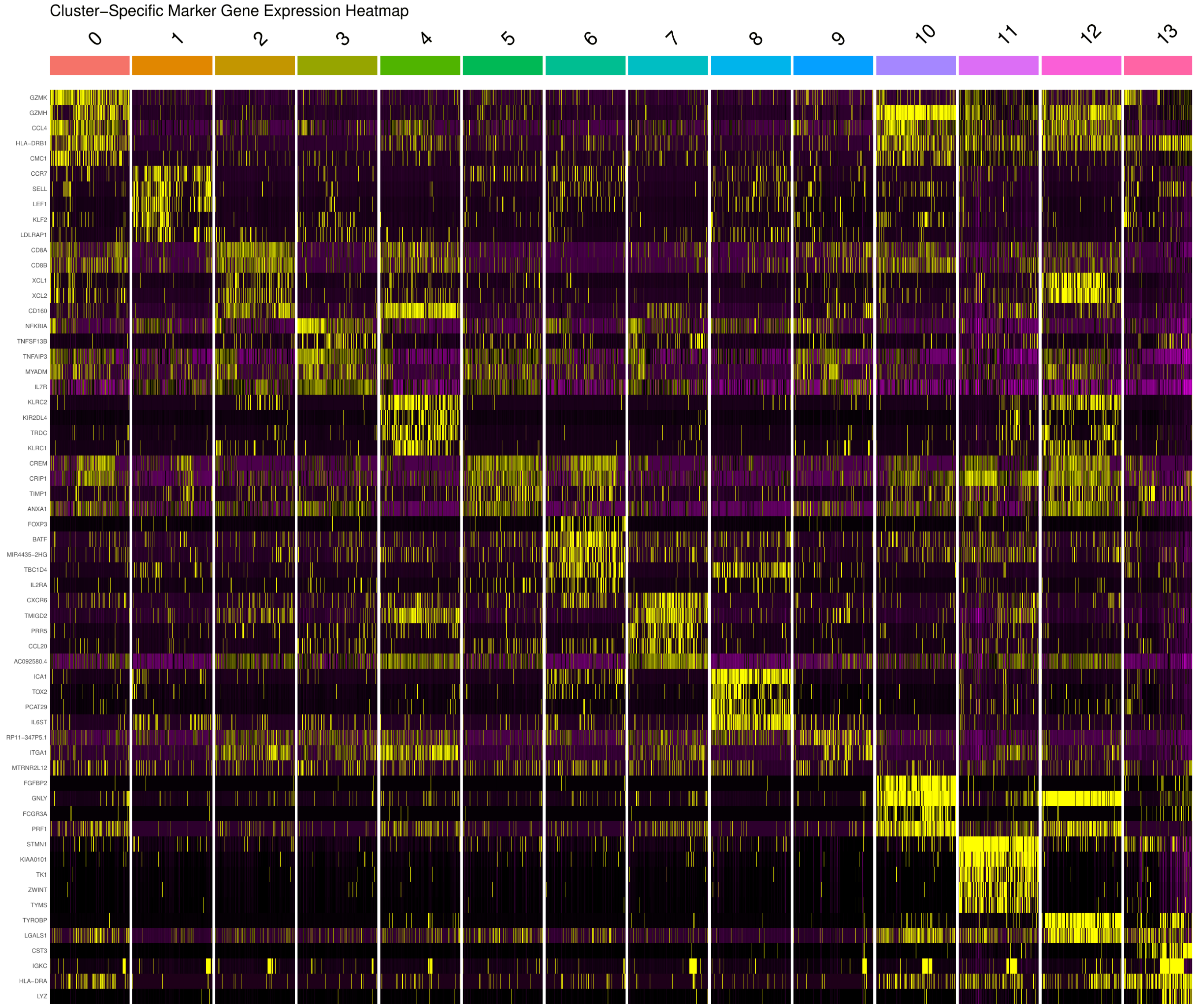
A total of **14** transcriptionally distinct clusters were identified.

Cluster visualization



Marker gene identification

Cluster-specific marker genes were identified to support biological interpretation.



7. Cell-Type Annotation

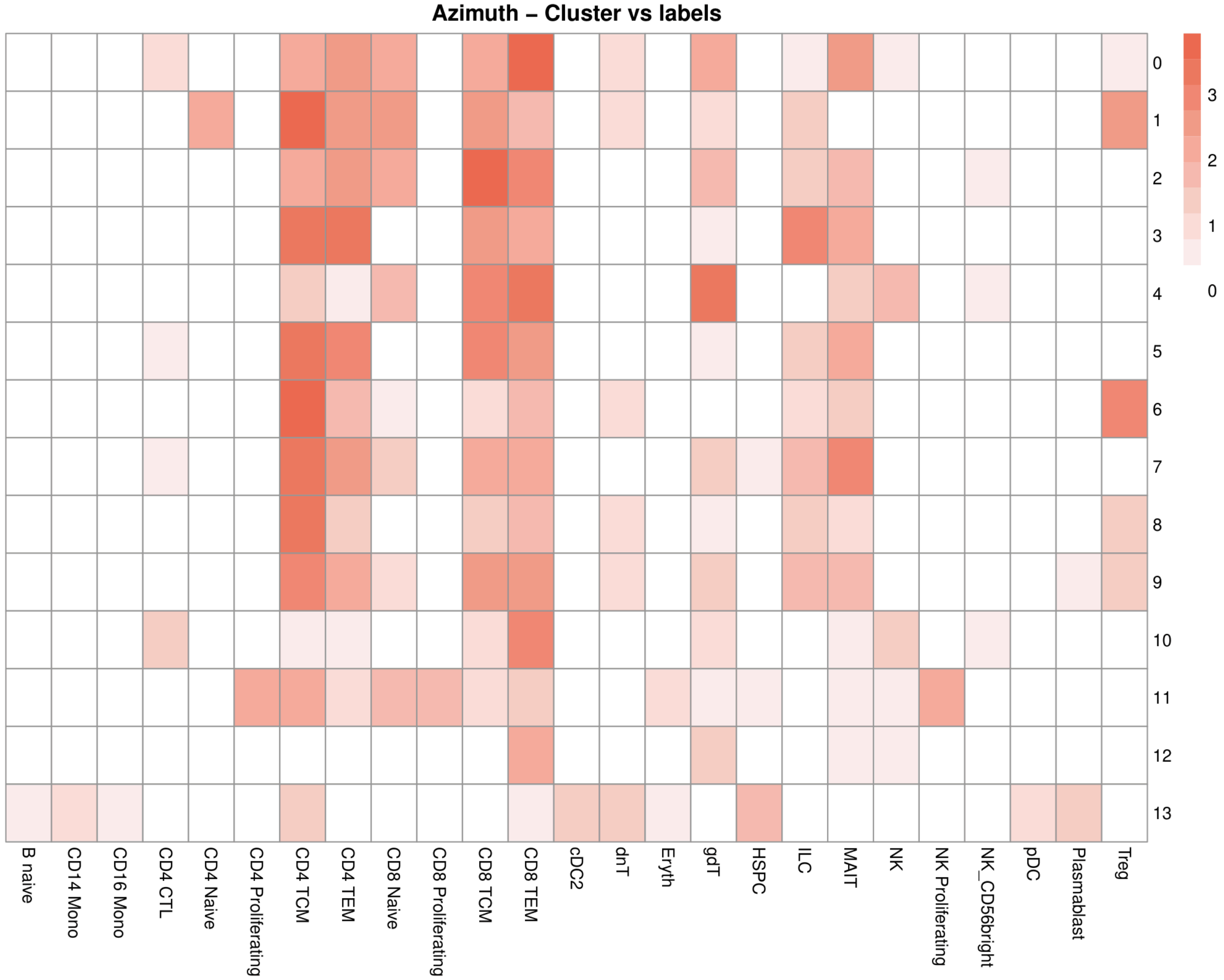
Cell identities were assigned using a multi-step consensus approach.

Cell identity assignment was based on a consensus approach combining reference-based annotation and marker-driven biological validation.

7.1 Reference-based annotation

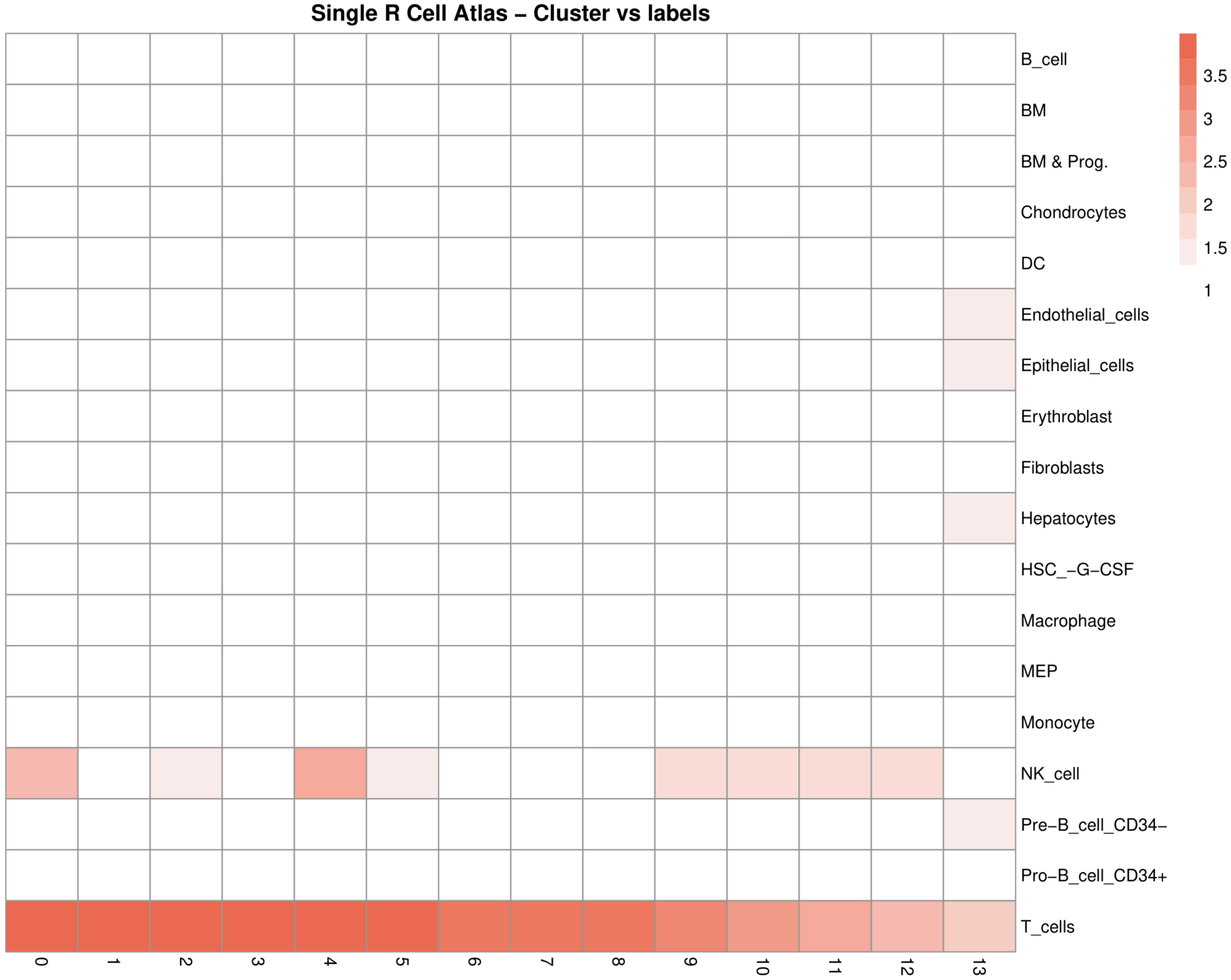
Azimuth

Reference mapping was performed using Azimuth to obtain predicted cell identities.



SingleR

SingleR annotation was performed using reference transcriptomic profiles.



7.2 Biological validation

Predicted identities were evaluated using canonical marker genes cluster-specific differential expression known immune and intestinal cell signatures.

7.3 Consensus integration

Integration of computational predictions and biological validation resulted in:

14 transcriptional clusters → **6** major cell populations

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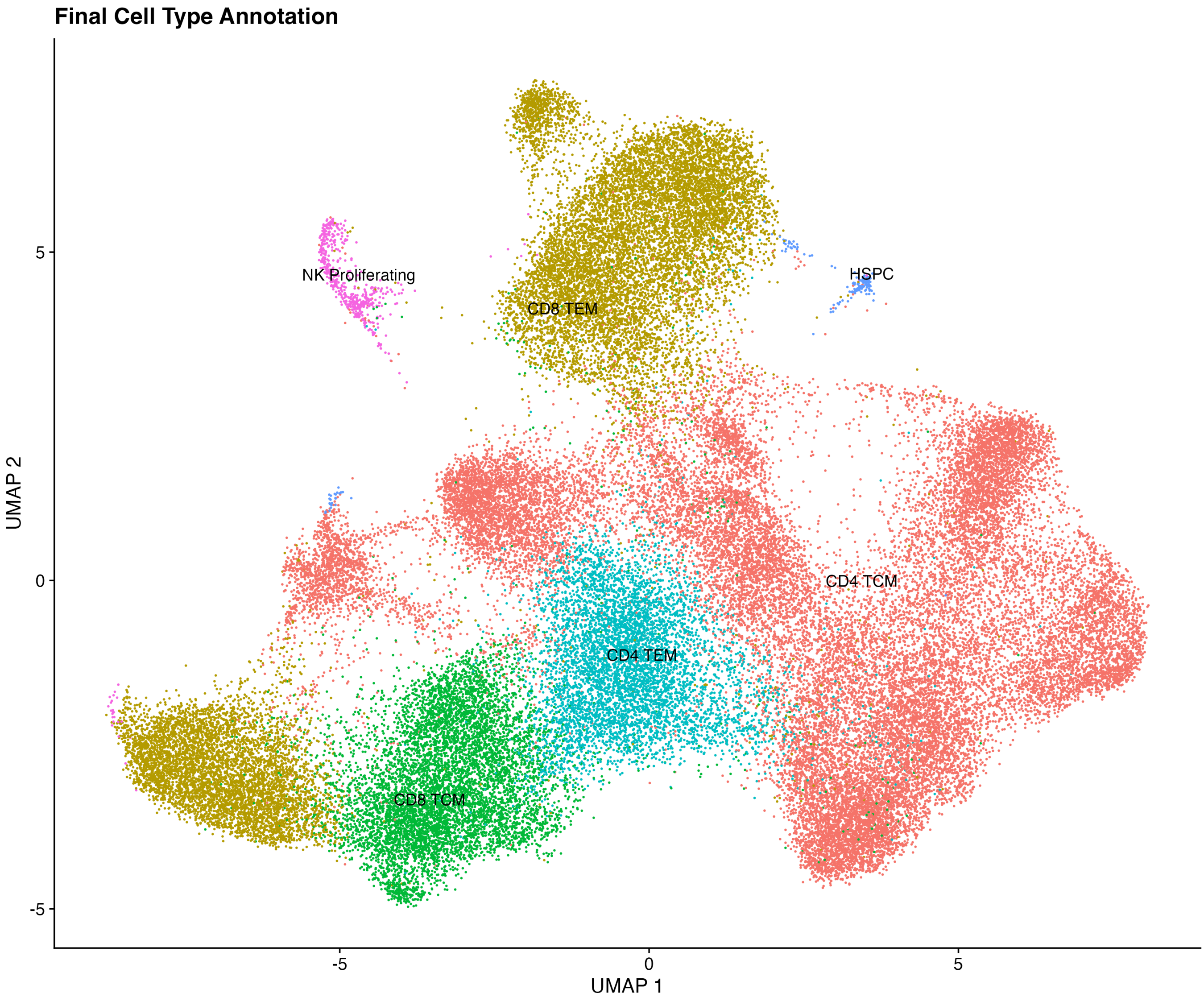
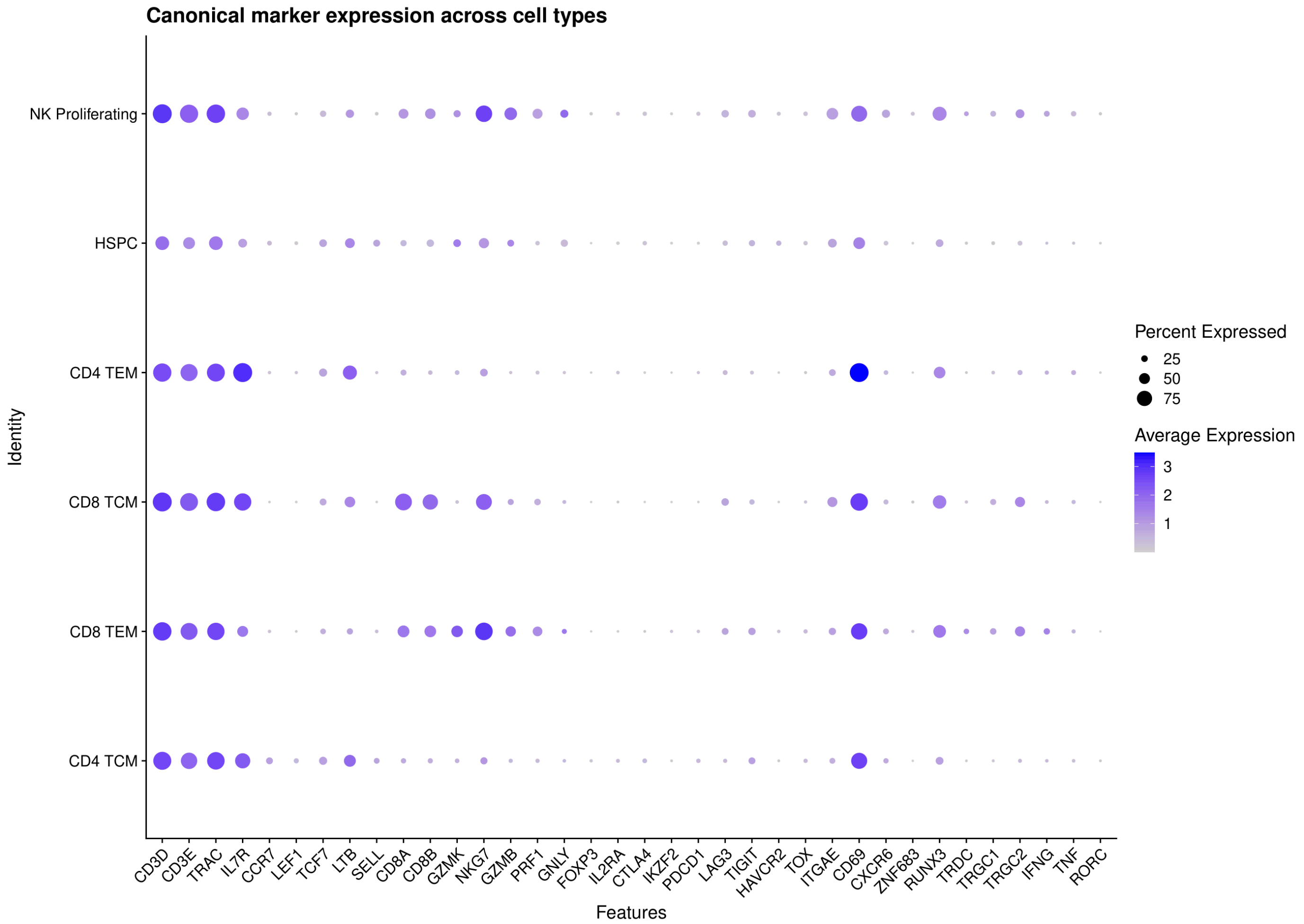
8. Final annotated cell atlas

Annotated UMAP



Cell-type validation and composition



9. Biological Interpretation

The integrated single-cell analysis framework enables the resolution of complex cellular ecosystems by combining unsupervised clustering, reference-based annotation, and marker-driven validation.

Graph-based clustering identified multiple transcriptionally distinct cell clusters in the integrated latent space, reflecting underlying heterogeneity within the dataset.

This structure provides the basis for downstream annotation and interpretation, where clusters are progressively mapped to biologically meaningful cell identities through consensus-based approaches.

Overall, this multi-layered strategy supports a reproducible and scalable framework for interpreting high-dimensional single-cell transcriptomic data.

10. Reproducibility

All analyses were performed in a standardized R environment to ensure reproducibility of results.

Software environment:

- R version: 4.6.0
- Seurat: 5.5.0
- Harmony: 2.0.3
- SingleR: 2.14.0
- Azimuth: 0.5.1
- celldex: 1.22.0

A complete session information log is stored in the `results/logs/` directory of the project repository, enabling full computational reproducibility of the analysis environment.

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11. Limitations and Considerations

Single-cell RNA-seq annotation represents a computational inference based on transcriptomic similarity and marker gene expression, rather than direct phenotypic validation.

Reference-based annotation methods may be influenced by:

composition of the reference dataset tissue-specific transcriptional variability disease-associated or context-specific cellular states

For this reason, cell identity assignment was based on a consensus approach integrating multiple annotation strategies, rather than relying on a single automated method.

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12. Deliverables

This analysis produces a complete set of reproducible outputs, including:

- quality-controlled Seurat object
- Harmony-integrated dataset
- graph-based clustering results
- marker gene tables for cluster characterization
- reference-based annotation outputs
- consensus cell-type classification
- publication-ready visualization figures

The final annotated dataset can support downstream analyses including:

1. compartment-specific subsetting
2. differential gene expression analysis
3. pathway and enrichment analysis
4. cell-state and trajectory characterization

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13. Conclusion

This workflow provides a reproducible end-to-end framework for single-cell RNA-seq data processing and analysis.

The global analysis successfully integrated multiple intestinal samples, identified 14 transcriptionally distinct clusters, and generated a consensus annotation resolving 6 major cellular populations.

Overall, the resulting annotated dataset provides a structured reference for downstream biological investigation of intestinal cellular heterogeneity and disease-associated transcriptional programs.